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A Major Project Phase-1 Report on

“Design and analysis of 6t sram using cadence virtuous”

Submitted in partial fulfillment of the requirements for the award of the degree of

BACHELOR OF ENGINEERING

in

ELECTRONICS AND COMMUNICATION ENGINEERING

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CERTIFICATE

This is to certify that the project work entitled “**Design and analysis of 6t sram using cadence virtuous**” is carried out by **Jeevan J M (4JN23EC036), Mithun S V (4JN23EC049), Arjun R (4JN23EC016), Abhishek B P (4JN23EC002)**, the bonafide students of JNN College of Engineering, Shimoga in partial fulfillment for the award of **Bachelor of Engineering** in department of **Electronics and Communication Engineering** of the Visvesvaraya Technological University, Belagavi, during the year **2025-26**. It is certified that all the corrections/suggestions indicated for internal assessment have been incorporated in the report deposited in the departmental library. The project report has been approved as it satisfies the academic requirements in respect of project work prescribed for the said degree.

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ABSTRACT

Static Random Access Memory (SRAM) is one of the most important memory elements used in modern digital systems due to its high speed, low latency, and reliable performance. In this project, a 6-Transistor (6T) SRAM cell is designed and analyzed using the Cadence Virtuoso environment with GPDK 45 nm technology. The project mainly focuses on the schematic design, simulation, layout implementation, and performance analysis of the SRAM cell for low-power and high-speed memory applications.

Initially, the single-bit 6T SRAM cell is designed at the schematic level using CMOS transistors and simulated to verify read, write, and hold operations. The layout of the single-bit SRAM cell is then implemented by following CMOS layout design rules in 45 nm technology, and its correctness is verified through Design Rule Check (DRC) and Layout Versus Schematic (LVS) analysis. Important performance parameters such as power consumption, propagation delay, and stability are analyzed to evaluate the efficiency of the SRAM design.

To demonstrate scalability and practical memory operation, an 8-bit SRAM test circuit is developed and simulated using the designed SRAM cell. The test results confirm successful memory functionality with reliable read and write operations. The proposed design provides better area efficiency, reduced power consumption, and improved performance suitable for modern VLSI memory applications.

ACKNOWLEDGEMENTS

We would like to acknowledge the help and encouragement given by various people during the course of this project. We would be thankful to our beloved principal of J. N. N. College of Engineering **Dr. Y Vijaya Kumar** for providing excellent academic climate. The support provided by the college and departmental library is greatly acknowledged.

We would like to express our sincere gratitude to **Dr. Sathyanarayana S V**, Head of the Department of ECE and Dean R&D, J. N. N. College of Engineering, Shivamogga, for his kind support, guidance and encouragement throughout the course of this work. We would like to express our grateful thanks to project coordinator **Dr. Anil Kumar J**, Assistant Professor, Department of Electronics and Communication Engineering, J. N. N. College of Engineering, Shivamogga, for his kind support, guidance and encouragement throughout the course of this work.

We were deeply indebted and very grateful to the invaluable guidance given by **Mr Ajay Betur P**, Assistant Professor, Department of Electronics and Communication Engineering, J. N. N. College of Engineering, Shivamogga, during this project work.

We would like to thank all the teaching and non-teaching staff of Dept. of ECE for their kind co-operation during the course of the work. Finally, we are thankful to our parents and friends, who helped us in one way or the other throughout our project work.

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Chapter 1

Introduction

1.1 Introduction

In modern digital electronic systems, memory plays a vital role in storing and processing data efficiently. Among different semiconductor memories, Static Random Access Memory (SRAM) is widely used in microprocessors, cache memories, embedded systems, and portable electronic devices because of its high speed, low access time, and reliable operation. SRAM retains data as long as power is supplied and does not require periodic refreshing, making it faster compared to Dynamic Random Access Memory (DRAM).

With the continuous advancement of Very Large Scale Integration (VLSI) technology, the demand for low-power, high-speed, and area-efficient memory circuits has significantly increased. The scaling of CMOS technology into nanometer dimensions has enabled the development of compact and high-performance memory architectures. In this project, a 6-Transistor (6T) SRAM cell is designed and analyzed using Cadence Virtuoso with GPDK 45 nm technology. The 6T SRAM cell is one of the most commonly used SRAM architectures because it provides a good balance between stability, speed, and power consumption.

The basic 6T SRAM cell consists of two cross-coupled CMOS inverters forming a bistable latch and two access transistors used for read and write operations. The SRAM cell operates in three modes: read, write, and hold. Proper transistor sizing and layout optimization are essential to ensure stable operation and improved performance parameters such as Static Noise Margin (SNM), power dissipation, and delay.

The main objective of this project is to design the schematic and layout of a single-bit 6T SRAM cell and analyze its functionality using Cadence Virtuoso tools. The schematic design is simulated to verify the correct read and write operations. The physical layout of the SRAM cell is created following CMOS design rules in 45 nm technology, and verification processes such as Design Rule Check (DRC) and Layout Versus Schematic (LVS) are performed to ensure correctness of the layout.

To demonstrate practical memory implementation, an 8-bit SRAM test circuit is developed using the designed SRAM cell. Although only a single-bit SRAM cell is physically designed and laid out, the test circuit simulations verify the operation of an 8-bit memory structure. The project also focuses on analyzing important parameters including power consumption, delay, and area optimization.

This work provides practical exposure to SRAM architecture, CMOS circuit design, layout techniques, and VLSI verification methodologies. The project helps in understanding the complete flow of memory design from schematic entry to layout verification using advanced semiconductor design tools and modern nanometer technology.

1.2 Problem Statement

With the continuous advancement of VLSI technology, modern electronic systems require memory circuits that provide high speed, low power consumption, improved stability, and reduced chip area. SRAM is widely used in processors, cache memory, and embedded systems because of its fast operation and reliable data storage capability. However, as CMOS technology scales down to nanometer dimensions, SRAM design faces several challenges such as increased leakage power, reduced stability, higher process variations, and performance degradation due to parasitic effects.

The main problem addressed in this project is the design and analysis of an efficient 6T SRAM cell using GPDK 45 nm technology in Cadence Virtuoso. The project focuses on developing a transistor-level SRAM design with proper read, write, and hold functionality while minimizing power consumption and delay. In addition, the project aims to implement the physical layout of a single SRAM cell, perform parasitic extraction and post-layout analysis, and verify the functionality of an 8-bit SRAM test circuit. The objective is to achieve a compact, stable, and efficient SRAM design suitable for modern VLSI memory applications.

1.3 Proposed Solution

The proposed solution for this project is the design and analysis of a 6T SRAM cell using Cadence Virtuoso with GPDK 45 nm CMOS technology to achieve a compact, low-power, and high-speed memory cell suitable for modern VLSI applications. The design uses six MOS transistors, where two cross-coupled CMOS inverters form a bistable latch for data storage and two access transistors control the read and write operations through bit lines and word lines. Proper transistor sizing and optimized circuit design techniques are used to improve stability, reduce power dissipation, and ensure reliable SRAM operation.

The project further proposes the implementation of the physical layout of a single 6T SRAM cell by following standard CMOS layout rules and performing verification using DRC and LVS analysis. To analyze practical circuit behavior, parasitic extraction and post-layout simulations are carried out to study the impact of resistance and capacitance on SRAM performance. An 8-bit SRAM test circuit is also developed at the schematic level to demonstrate large-scale memory operation. The overall solution focuses on achieving better power efficiency, reduced delay, improved stability, and optimized area for advanced memory applications in VLSI systems.

1.4 Proposed Solution

The proposed project presents the design and analysis of a high-speed and low-power 6T SRAM cell using Cadence Virtuoso with GPDK 45 nm CMOS technology. The system is designed to achieve stable read, write, and hold operations while reducing power consumption, delay, and silicon area. The proposed solution mainly focuses on transistor-level design, schematic simulation, layout implementation, parasitic extraction, and post-layout analysis of the SRAM cell.

The proposed SRAM architecture uses six MOS transistors, where two cross-coupled CMOS inverters form a bistable latch for storing one bit of data, and two NMOS access transistors are used to control read and write operations through bit lines and word lines. The design methodology combines circuit-level optimization and layout-level verification to achieve efficient SRAM performance for modern VLSI applications.

The proposed solution integrates the following stages:

- Transistor-level SRAM cell design
- Schematic simulation and functional verification
- Layout implementation using CMOS design rules
- DRC and LVS verification
- Parasitic extraction and post-layout simulation
- Performance analysis of SRAM parameters

Initially, the 6T SRAM cell is designed at the transistor level using PMOS and NMOS transistors available in GPDK45 technology. The SRAM cell contains two cross-coupled inverters which maintain the stored logic value and two access transistors controlled by the

word line signal. Proper transistor sizing is performed to improve stability, reduce leakage power, and ensure reliable switching during read and write operations.

After completing the transistor-level design, the schematic simulation stage is carried out to verify the functionality of the SRAM cell. The following SRAM operating modes are analyzed:

1. **Write Operation:** During the write operation, the bit lines are driven with the required input data while the word line is activated. The access transistors allow the new data to overwrite the previously stored value inside the SRAM cell.
2. **Read Operation:** During the read operation, the stored data inside the SRAM cell is transferred to the bit lines when the word line is enabled. The read stability and correct data sensing are analyzed through transient simulation.
3. **Hold Operation:** In hold mode, the word line remains disabled and the cross-coupled inverters retain the stored data without refreshing. This operation verifies the memory retention capability of the SRAM cell.

To demonstrate practical memory implementation, an 8-bit SRAM test circuit is developed at the schematic level by connecting multiple SRAM cells. The test circuit is simulated using transient analysis to verify correct multi-bit read and write functionality.

After schematic verification, the physical layout of a single 6T SRAM cell is implemented using CMOS layout design techniques. Different layers such as diffusion, polysilicon, metal, contacts, and wells are carefully placed and routed according to GPDK45 design rules. Symmetrical layout design is followed to minimize area and improve matching characteristics.

The completed layout is then verified using the following verification techniques:

1. **Design Rule Check (DRC):** DRC is performed to ensure that the layout satisfies all fabrication design rules including spacing, width, and overlap constraints.
2. **Layout Versus Schematic (LVS):** LVS verification is carried out to confirm that the generated layout is electrically equivalent to the designed schematic.

Once the layout verification is completed, parasitic extraction is performed to obtain the parasitic resistance and capacitance introduced during physical implementation. These parasitic components affect the practical performance of the SRAM cell.

The extracted parasitic netlist is then used for post-layout simulation. Post-layout analysis helps evaluate the real-time SRAM behavior by considering interconnect delays and parasitic effects. The obtained results are compared with schematic simulation results to study performance variations.

Finally, important SRAM performance parameters are evaluated including:

- Power Consumption
- Propagation Delay
- Read and Write Stability
- Area Efficiency
- Signal Integrity
- Access Time

Noise residual analysis identifies unnatural smoothing and editing artifacts introduced during image manipulation. Edge detection highlights discontinuities in object boundaries, while Laplacian filtering detects abrupt pixel intensity changes. Local variance analysis helps identify texture inconsistencies within manipulated regions.

The extracted spatial and frequency features are then passed to a **Vision Transformer (ViT)** based deep learning model. The Vision Transformer learns complex image representations and predicts the probability of the image being AI-generated or manipulated.

The system contains multiple specialized classification models including:

- Real Image Detection Model
- Deepfake Detection Model
- Tampered Image Detection Model
- Masked Image Detection Model
- AI-Generated Image Detection Model

The proposed solution provides a compact, reliable, and efficient SRAM design suitable for modern low-power VLSI memory applications and gives practical exposure to complete SRAM design flow from schematic design to post-layout analysis.

1.5 Literature Survey

1.5.1 Design and Analysis of CMOS based 6T SRAM Cell at Different Technology Nodes

Authors: K. Agarwal et al.

Year of Publication: 2020

Publication: IOP Publishing Ltd

The paper analyzes the performance of a 6T SRAM cell at different CMOS technology nodes by evaluating parameters such as power consumption, delay, and Static Noise Margin (SNM).

The study shows that scaling down the technology node improves SRAM performance, especially in terms of speed, stability, and reduced delay.

Improved speed and stability, reduced delay, better SRAM efficiency at smaller technology nodes. Increased fabrication complexity and leakage issues in lower technology nodes.

1.5.2 Performance Evaluation of 6T SRAM in Sub-3nm CFET Technology

Authors: Anirban Kar, Mahdi Benkhelifa et al.

Year of Publication: 2025

Publication: 9th IEEE Electron Device Technology and Manufacturing Conference.

The proposed work evaluates a 6T SRAM cell using advanced Sub-3nm CFET technology to improve memory density, scalability, and performance.

The results demonstrate that CFET technology provides better scalability, higher efficiency, and improved memory density compared to traditional CMOS technology.

Higher scalability, better efficiency, improved memory density and performance. Advanced fabrication complexity and high implementation cost.

1.5.3 Parametric Inspection of 6T SRAM Cell

Authors: Prince Kumar Pandey et al

Year of Publication: 2021

Publication: International Conference on Trends in Electronics and information

This paper performs a parametric analysis of the 6T SRAM cell by studying SNM, power consumption, and delay characteristics.

The study identifies important factors affecting SRAM stability and overall performance.

Better understanding of SRAM parameters and stability improvement techniques. Performance variations due to process and environmental changes.

1.5.4 Low Voltage 6T SRAM Cell Design Using 90nm and 45nm CMOS Technology

Authors: T. Kalpana et al.

Year of Publication: 2024

Publication: 7th International Conference on Device and System

The paper designs and analyzes a low-voltage 6T SRAM cell using 90nm and 45nm CMOS technologies and compares their performance

Results show that lower technology nodes help reduce power consumption while improving SRAM efficiency.

Reduced power consumption, improved low-voltage operation, better efficiency Reduced noise margin and increased sensitivity at low voltages.

1.5.5 Performance Analysis of 6T SRAM Cell and 64×64 Memory Array

Authors: Tushar Singh et al.

Year of Publication: 2023

Publication: 1st International Conference on Circuits, Power and Intelligent Systems .

This paper analyzes a 6T SRAM cell integrated with a 64×64 memory array focusing on scalability and power efficiency.

The study highlights the importance of optimization techniques for large-scale SRAM array implementation.

Good scalability, suitable for large memory arrays, improved optimization. Increased design complexity and higher area requirement for large arrays.

1.6 Objectives

The main objectives of the project are:

1. design 6T SRAM cell at transistor level
2. To implement the layout of a single 6T SRAM cel
3. To perform schematic simulation of an 8-bit SRAM
4. To extract parasitic parameters and carry out post-layout simulation
5. To evaluate key SRAM performance metrics.

1.7 Methodology

1. Design of 6T SRAM Cell at Transistor Level:

The first step of the project is the transistor-level design of the 6T SRAM cell using Cadence Virtuoso in GPDK 45 nm technology. The SRAM cell consists of six MOS transistors, including two PMOS transistors, two NMOS driver transistors, and two NMOS access transistors. Initially, the required design library and technology files are created in Cadence Virtuoso. The schematic of the SRAM cell is then developed using CMOS transistor components available in the GPDK45 library.

Two cross-coupled CMOS inverters are connected to form a bistable latch capable of storing one bit of data. Two access transistors are connected between the storage nodes and bit lines to control read and write operations using the word line signal. Proper transistor sizing is selected to achieve good read stability, write ability, and low power consumption. The designed schematic is checked for connectivity and electrical errors before proceeding to simulation.

2. Schematic Simulation of 8-Bit SRAM :

After designing the single-bit SRAM cell, schematic simulations are carried out to verify its functionality. Initially, simulations are performed for the basic operations such as hold, read, and write modes using transient analysis. Input signals including word line, bit line, and bit bar line are applied to verify correct memory operation.

To demonstrate practical memory implementation, an 8-bit SRAM test circuit is developed by connecting multiple SRAM cells at the schematic level. The simulation setup includes pulse voltage sources, control signals, and load conditions. Transient simulations are performed using Cadence Spectre simulator to observe the timing

behavior and data transfer operations of the 8-bit SRAM. The output waveforms are analyzed to confirm successful read and write functionality.

3. Layout Implementation of Single 6T SRAM Cell:

Once the schematic functionality is verified, the physical layout of the single 6T SRAM cell is designed. The layout is created using layers such as diffusion, polysilicon, metal, and contacts according to GPDK45 CMOS design rules. PMOS transistors are placed in the N-well region while NMOS transistors are placed in the P-substrate region.

Careful placement and routing techniques are followed to minimize area and improve symmetry of the SRAM cell. Power supply lines (VDD and GND), bit lines, and word lines are routed properly to ensure correct operation. After layout completion, Design Rule Check (DRC) is performed to verify that the layout satisfies all fabrication design rules. Layout Versus Schematic (LVS) verification is then carried out to confirm that the physical layout matches the original schematic design.

4. Parasitic Extraction and Post-Layout Simulation:

After successful DRC and LVS verification, parasitic extraction is performed on the layout. During this process, parasitic resistance and capacitance values introduced by interconnections and routing are extracted from the physical layout. These parasitic effects influence the real-time performance of the SRAM cell.

The extracted netlist is then used for post-layout simulation. Transient analysis is again performed to study the practical behavior of the SRAM cell under parasitic conditions. The output waveforms obtained from post-layout simulation are compared with schematic simulation results to analyze variations in delay, power consumption, and signal integrity caused by parasitic components.

5. Evaluation of SRAM Performance Parameters:

Finally, the performance of the designed SRAM cell is evaluated based on several important parameters. Power consumption is measured during read, write, and hold operations to analyze energy efficiency. Propagation delay and access time are calculated to evaluate the speed performance of the SRAM cell.

The stability of the SRAM cell is analyzed by observing correct data retention and switching behavior during operations. Area efficiency of the layout is also examined to determine compactness of the design. The obtained results are analyzed and compared to verify that the designed 6T SRAM cell satisfies the requirements of modern low-power and high-speed VLSI memory applications.

1.8 Applications

The proposed system has applications in several domains including:

- Cache memory in microprocessors and microcontrollers
- High-speed memory units in embedded systems
- Portable electronic devices such as smartphones and tablets
- Low-power VLSI memory applications
- Digital signal processing (DSP) systems
- Communication and networking devices

1.9 Limitations

The proposed system has certain limitations:

- SRAM occupies larger chip area due to the use of six transistors per memory cell, which increases fabrication cost.
- Power consumption is higher compared to DRAM, especially in large memory arrays.
- SRAM density is lower, making it unsuitable for very high-capacity memory applications.
- As technology scales down, SRAM faces stability issues such as reduced Static Noise Margin (SNM).

1.10 Organisation of the Project

The project report is organized into four chapters.

Chapter 1 introduces the problem statement, proposed solution, literature survey, objectives, methodology, applications, and limitations of the proposed system.

Chapter 2 explains the design and implementation details of the hybrid forensic framework, including system architecture, block diagrams, and implementation procedures.

Chapter 3 presents the results obtained from different analysis techniques and discusses the performance of the proposed system.

Chapter 4 concludes the project and discusses future enhancements and improvements.

Chapter 2

Design and Implementation

2.1 Introduction

This chapter explains the design and implementation details of the proposed hybrid image forensic system titled **“Design and Analysis of 6T SRAM using Cadence virtouso”**. SRAM (Static Random Access Memory) is one of the most important semiconductor memories used in modern digital systems due to its high speed, low power consumption, and reliable performance. It is widely used in cache memory, processors, embedded systems, and portable electronic devices. Unlike DRAM, SRAM does not require periodic refreshing to retain data, making it faster and more efficient for high-speed applications. With the rapid growth of VLSI technology and scaling of CMOS devices into nanometer dimensions, designing compact and efficient SRAM cells has become an important area of research in memory circuit design.

2.2 System Design

The proposed SRAM system architecture consists of a 6T SRAM cell, control lines, bit lines, sense circuitry, and an 8-bit SRAM test structure designed using Cadence Virtuoso with GPDK 45 nm technology. The architecture mainly performs three operations: read, write, and hold.

The basic 6T SRAM cell contains:

- Two cross-coupled CMOS inverters for storing one bit of data
- Two NMOS access transistors controlled by the Word Line (WL)
- Two Bit Lines (BL and BLB) used for data transfer

During the write operation, data is applied through BL and BLB while the WL is enabled. During the read operation, the stored data is sensed through the bit lines. In hold mode, the WL remains disabled, allowing the cross-coupled inverters to retain the stored data.

For demonstrating multi-bit memory functionality, multiple SRAM cells are connected to form an 8-bit SRAM test circuit. The overall architecture includes:

- 6T SRAM memory cell
- Word Line driver
- Bit Line and Bit Bar Line
- Precharge circuit
- Sense amplifier
- Sense amplifier
- Output data sensing block

The complete architecture is verified through schematic simulation, layout implementation of a single SRAM cell, parasitic extraction, and post-layout analysis to evaluate SRAM performance parameters such as power, delay, and stability.

2.3 Flow Chart

The project starts with designing a 6T SRAM cell using Cadence Virtuoso and GPDK 45 nm technology. The schematic is created using six MOS transistors and simulated to verify read, write, and hold operations. Proper transistor sizing is done to achieve stable and efficient SRAM performance.

After schematic verification, an 8-bit SRAM test circuit is simulated to check multi-bit memory operation. Then, the layout of a single 6T SRAM cell is designed and verified using DRC and LVS checks. Parasitic extraction and post-layout simulation are performed to analyze practical SRAM behavior. Finally, parameters such as power, delay, stability, and area are evaluated.

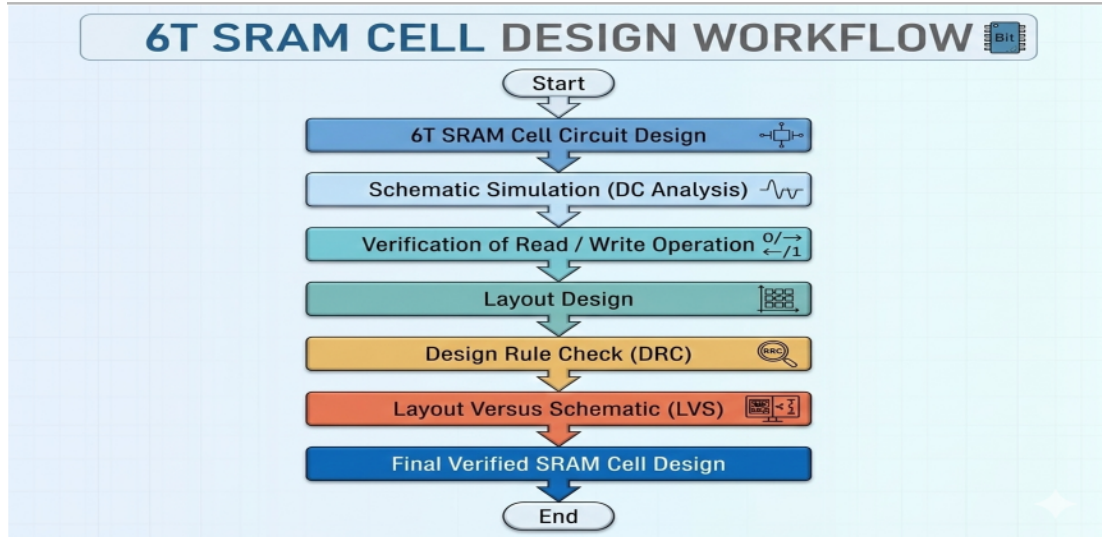


Figure 2.1: Flow Chart of Designing the 6 transistor Static RAM

2.4 Read Operation

In the read operation, the stored data inside the SRAM cell is accessed through the bit lines without changing the stored value. Before starting the read process, both bit lines BL and BLB are precharged to logic HIGH (VDD). Initially, the Word Line (WL) is kept LOW, which keeps the access transistors OFF and isolates the SRAM cell from the bit lines.

To perform the read operation, the Word Line is activated by making WL HIGH. This turns ON the two access transistors and connects the internal storage nodes of the SRAM cell to BL and BLB. Assume that the SRAM cell stores logic '1' at node Q and logic '0' at node QB. Since QB is connected to logic LOW, the corresponding bit line slightly discharges through the NMOS path, while the other bit line remains near VDD.

A sense amplifier detects the small voltage difference between BL and BLB and determines the stored data. During the read operation, proper transistor sizing is important because the stored data should not flip accidentally. Therefore, the pull-down NMOS transistors inside the SRAM cell are designed stronger than the access transistors to maintain read stability.

2.5 Write Operation

In the write operation, new data is stored into the SRAM cell through the bit lines. The data to be written is first applied to BL and BLB in complementary form. For example:

- To write logic '1': BL = 1 and BLB = 0
- To write logic '0': BL = 0 and BLB = 1

Initially, the Word Line is LOW, so the SRAM cell remains isolated from the bit lines. When the Word Line is made HIGH, the access transistors turn ON and connect the bit lines to the internal storage nodes of the SRAM cell.

Suppose the cell initially stores logic '0' and a logic '1' needs to be written. The voltage applied on the bit lines forces one internal node to VDD and the other node to GND. Due to the cross-coupled inverter structure, the SRAM cell quickly changes its state and stores the new data permanently as long as power is supplied.

During the write operation, the access transistors and write drivers must be strong enough to overwrite the previous stored value inside the SRAM cell. After successful data writing, the Word Line is turned LOW again, disconnecting the SRAM cell from the bit lines and retaining the newly stored data.

2.6 Summary

This project focuses on the design and analysis of a 6T SRAM (Static Random Access Memory) cell using Cadence Virtuoso with GPDK 45 nm CMOS technology. SRAM is an important memory element widely used in processors, cache memory, embedded systems, and portable electronic devices because of its high-speed operation and reliable data storage capability. The main objective of this project is to develop a compact, stable, and low-power SRAM cell suitable for modern VLSI applications.

The project begins with the transistor-level design of the 6T SRAM cell using six MOS transistors. Two cross-coupled CMOS inverters are used to form a bistable latch capable of storing one bit of data, while two NMOS access transistors control the read and write operations through the word line and bit lines. Proper transistor sizing is carried out to improve stability, reduce leakage power, and achieve efficient SRAM operation.

After completing the schematic design, simulations are performed to verify the three major SRAM operations: read, write, and hold. During the write operation, data is applied through the bit lines and stored inside the SRAM cell by enabling the word line. During the read operation, the stored data is sensed through the bit lines without disturbing the stored value. In hold mode, the SRAM cell retains the stored data without requiring any refresh operation. The transient simulation results confirm the correct functionality of the designed SRAM cell.

The project also includes the physical layout implementation of a single 6T SRAM cell using CMOS layout design techniques. Layout layers such as diffusion, polysilicon, metal, contacts, and wells are carefully designed according to GPDK45 technology rules. After layout completion, Design Rule Check (DRC) is performed to verify that the layout satisfies fabrication constraints, while Layout Versus Schematic (LVS) verification ensures that the layout matches the original schematic design.

After successful layout verification, parasitic extraction is carried out to obtain the parasitic resistance and capacitance introduced during physical implementation. Post-layout simulations are then performed using the extracted netlist to analyze the practical behavior of the SRAM cell under real-time conditions. The post-layout results are compared with schematic simulation results to study the effect of parasitic components on SRAM performance.

Overall, this project provides complete practical exposure to SRAM design methodology, CMOS circuit design, layout implementation, verification techniques, parasitic analysis, and post-layout simulation using advanced VLSI design tools. It also helps in understanding the complete memory design flow from schematic creation to physical implementation and performance evaluation.

Chapter 3

Results and Discussion

3.1 Results for Objective 1

The first objective of the project was successfully completed by designing a 6T SRAM cell at the transistor level using Cadence Virtuoso with GPDK 45 nm CMOS technology. The SRAM cell was designed using six MOS transistors, where two cross-coupled CMOS inverters act as a bistable latch for storing one bit of data and two access transistors control the read and write operations. Proper transistor sizing and circuit connections were implemented to achieve stable operation, low power consumption, and reliable SRAM functionality.

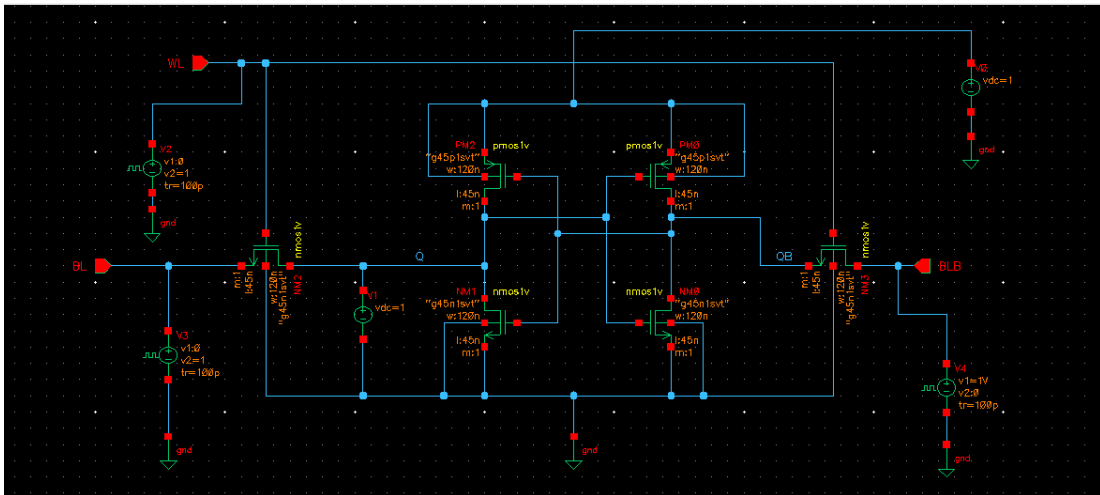


Figure 3.1: Schematic Design of 6 transistor Static RAM

Mainly in this objective we focused on the design the cmos schematic of the 6 transistor static random access memory . Generally we use only 2 pmos transistor for pull up of the circuit . And 4 nmos transistor are used there 2 nmos are used to wl and bl line and other 2 transistor is for pull down of the circuit. Two cross coupled inverter are used .

3.2 Results for Objective 2

The second objective of the project involves the physical layout implementation of a single 6T SRAM cell using Cadence Virtuoso with GPDK 45 nm CMOS technology. The layout is created by placing PMOS and NMOS transistors and routing the required connections using diffusion, polysilicon, metal, and contact layers according to CMOS fabrication design rules. Proper layout symmetry and area optimization techniques are followed to achieve compact and efficient design. After layout completion, Design Rule Check (DRC) and Layout Versus Schematic (LVS) verification are performed to ensure that the layout is error-free and electrically equivalent to the schematic design.

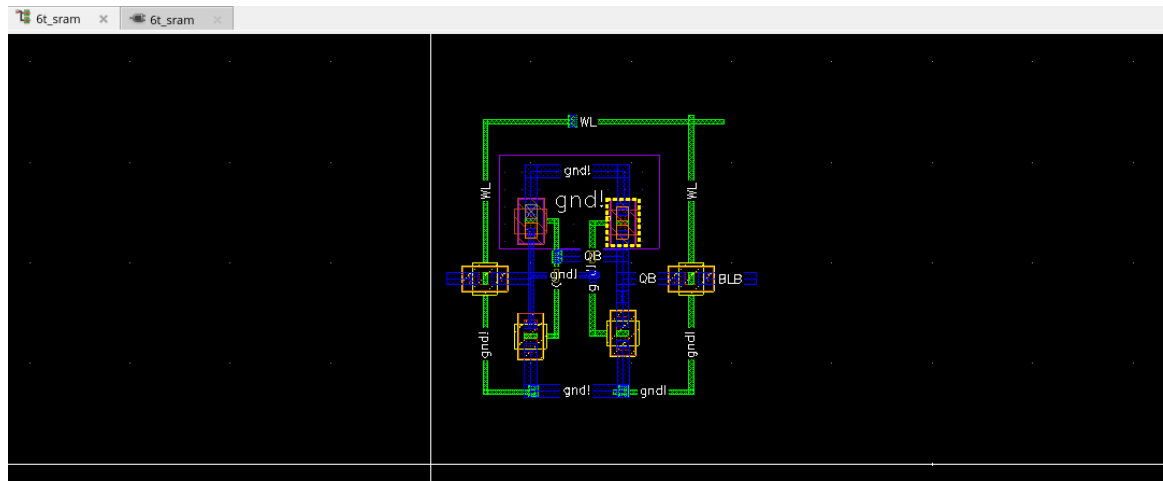


Figure 3.2: Layout Design of 6 transistor Static RAM

In the second objective we mainly focused on the layout design the cmos circuit of 6t sram . Layout is design considering the all design factor for effcient of the system . To achieve the compact design the system . In layout both DRC(design rule check) and LVS(layout vs schematic) are verified and 0 error is achieved .

Chapter 4

Conclusion

The project titled “**Design and Analysis of 6T SRAM using Cadence Virtuoso**” In this project, the design and physical implementation of a 6T SRAM cell were successfully carried out using Cadence Virtuoso with GPDK 45 nm CMOS technology. The first objective of the project, which involved designing the 6T SRAM cell at the transistor level, was successfully completed. The SRAM cell was designed using six MOS transistors consisting of two cross-coupled CMOS inverters and two access transistors. Proper transistor sizing and circuit connections were implemented to achieve stable read, write, and hold functionality with reduced power consumption and improved reliability. The schematic design confirmed the fundamental operation of the SRAM cell suitable for modern low-power VLSI memory applications.

The Second objective, which focused on the layout implementation of the single 6T SRAM cell, was also successfully completed. The physical layout was designed using CMOS layout layers such as diffusion, polysilicon, metal, contacts, and wells according to GPDK45 design rules. Proper placement and routing techniques were followed to obtain a compact and symmetrical layout structure. Verification processes including Design Rule Check (DRC) and Layout Versus Schematic (LVS) were performed to ensure that the layout was error-free and electrically equivalent to the schematic design. The successful completion of these objectives provided practical exposure to transistor-level SRAM design, CMOS layout implementation, and VLSI verification methodologies, forming a strong foundation for further SRAM performance analysis and advanced memory design applications.

4.1 References

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